

Two-Dimensional Ising Model Simulation Report

1. Introduction

This project investigates the two-dimensional Ising model using Monte Carlo simulation with the Metropolis algorithm. The temperature dependence of energy, magnetization, and magnetic susceptibility is studied. Finite-size effects are analyzed by comparing systems with different lattice sizes, and spin configurations are visualized at different temperatures to observe the phase transition behavior.

2. Part 1: Basic Ising Model Simulation

2.1 Energy versus Temperature

The energy-temperature curve shows that the system energy changes continuously with temperature. At low temperature, spins tend to align, producing a low-energy ordered state. As temperature increases, thermal fluctuations become stronger and the energy increases because more disordered spin configurations become accessible.

2.2 Magnetization versus Temperature

The magnetization remains close to one at low temperature, indicating a ferromagnetic ordered phase. Around the critical region, magnetization decreases rapidly. At high temperature, magnetization approaches zero, showing that the system enters a disordered paramagnetic phase.

2.3 Spin Configuration

The spin configuration at low temperature shows large domains of aligned spins. This demonstrates that thermal energy is insufficient to destroy the magnetic order. With increasing temperature, the spin pattern becomes more random due to stronger thermal fluctuations.

3. Part 2: Finite Size Effect and Critical Temperature

3.1 Comparison Between $L=32$ and $L=64$

The simulations were performed for two lattice sizes, $L=32$ and $L=64$. The magnetization curves show similar phase transition behavior, while the larger lattice produces a smoother transition because finite-size fluctuations are reduced.

3.2 Susceptibility and Critical Temperature

Magnetic susceptibility reaches a maximum near the critical temperature because the system exhibits strong fluctuations. The peak position provides an estimation of the critical temperature. The obtained values are close to the theoretical critical temperature of the two-dimensional Ising model.

4. Part 3: Critical Fluctuation Analysis

4.1 Magnetization Trace

The magnetization trace was analyzed near and above the critical temperature. At $T=2.25$, close to the transition point, the magnetization shows strong fluctuations because the correlation length becomes large. At $T=2.50$, above the critical temperature, the magnetization fluctuates around zero, indicating the absence of long-range magnetic order.

4.2 Spin Configuration Snapshots

Spin configurations were obtained at $T=2.0$, $T=2.3$, and $T=2.5$. At $T=2.0$, the system remains ordered with large aligned spin regions. At $T=2.3$, near the critical temperature, complex domain structures appear due to critical fluctuations. At $T=2.5$, spins become randomly distributed, representing the disordered phase.

5. Conclusion

The Monte Carlo simulation successfully reproduces the phase transition behavior of the two-dimensional Ising model. The temperature dependence of energy, magnetization, and susceptibility confirms the transition from ferromagnetic order to a disordered phase. Finite-size analysis and spin configuration visualization further demonstrate critical fluctuations near the transition temperature.

Figures generated from simulation results should be inserted into the corresponding sections before submission.